

Monte Carlo Simulation of a Simple Polymer Chain

Aim:

The goal of this project is to implement a Monte Carlo (MC) simulation program to explore the conformational landscape of a linear polymer (polyethylene). The simulation must incorporate torsional energy rotations and intramolecular Lennard-Jones interactions. Students will coordinate all tasks required to produce a complete and functional parallel MC code.

Scientific goal:

Develop a modular, well-documented Monte Carlo program capable of simulating a polyethylene model in which each C–C bond can adopt a different torsional angle during the simulation. The initial version of the program should consider only carbon atoms (united-atom force field). In a second version, explicit hydrogens must be included.

The simulation must:

- Generate an initial polymer conformation.
- Apply Monte Carlo moves that modify torsional states.
- Compute the energy using a Lennard-Jones potential between non-bonded carbon atoms.
- Perform Metropolis acceptance/rejection steps.
- Output structural and energetic observables.
- Provide post-processing tools and visualization of results.

Parallelization requirements (MPI):

1. **Replica parallelism:** Multiple independent replicas.
2. **Parallel energy evaluation:** Distribution of the energy calculation across nodes.
3. **Mixed parallelism:** Several replicas, each using parallel energy evaluation.

Minimum required tools:

- Git / GitHub (documentation, branches, commits, issue tracking).
- Makefiles (compilation, execution pipelines, figure generation).
- MPI (parallel version of the program).
- Optional Python scripts for post-processing and plotting.

Deliverables

March 12th — Sequential Code Delivery

(To be submitted in Campus Virtual and GitHub)

Evaluation criteria:

- Code modularization.
 - Well-indented and well-documented code.
 - Proper documentation of the repository.
 - Effective use of Makefiles to:
 - build the executable
 - run simulations
 - generate figures
 - A *results* folder containing a simulation of a **500-carbon polymer at fixed temperature**, including:
 - Energy evolution
 - Torsion angle distributions
 - Radius of gyration and end-to-end distance
 - Monte Carlo trajectory
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April 12th — Project Report Delivery (24h)

(To be submitted in Campus Virtual and GitHub)

Memory format:

- **Text:**
Must include:
 - Explanation of all implemented algorithms
 - Scalability study of the parallel code
 - Test cases
 - Data analysis

The manuscript must be delivered as a **PDF document**.

- **Code:**
A zip or `tar.gz` file containing:
 - All source code
 - Makefiles
 - Instructions for compilation and execution

Every subroutine must indicate the author. If shared, individual responsibilities must be explicitly stated.

All code must be properly documented.

- **Deliverable:**
 - PDF project report
 - `.zip` file containing all code and instructions

Project Presentation (April 15th — To be confirmed)

Format: Prezi, PowerPoint, PDF or similar.

Responsibilities: Each student prepares their own slides, forming part of a single, integrated presentation.

Duration:

- 20–30 minutes total presentation
- 15–30 minutes of questions from the instructor

Evaluation Criteria

- Respect for all deadlines (late deliveries will be penalized).
- Correct and effective use of GitHub, Makefiles, and parallelization techniques in all deliverables.